



FINAL MARK

**GIRRAWEEEN HIGH SCHOOL
MATHEMATICS
YEAR 12 HSC TASK 2 2015
ANSWERS COVER SHEET**

Name: _____

QUESTION	MARK	H2	H3	H4	H5	H6	H7	H8	H9
Q1 – Q5	/5				✓				✓
Q6	/9				✓				✓
Q7	/13				✓	✓			✓
Q8	/11				✓				✓
Q9	/13				✓	✓			✓
Q10	/17				✓				
Q11	/9				✓			✓	
Q12	/13				✓				✓
TOTAL									
	/90				/90	/26		/12	/90

- H2 constructs arguments to prove and justify results.
- H3 manipulates algebraic expressions involving logarithmic and exponential functions.
- H4 expresses practical problems in mathematical terms based on simple given models.
- H5 applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems.
- H6 uses the derivative to determine the features of the graph of a function.
- H7 uses the features of a graph to deduce information about the derivative.
- H8 uses techniques of integration to calculate areas and volumes.
- H9 communicates using mathematical language, notation, diagrams and graphs.

GIRRAWEE HIGH SCHOOL

MATHEMATICS

Task 2

Year 12 Mathematics

2015

Time Allowed: 90 minutes

Instructions:

- There are 12 questions in this paper. All questions are compulsory.
- Start each question (6 – 12) on a new sheet of paper.
- Write on one side of the paper only.
- Show all necessary working.
- Board-approved calculators may be used.
- Marks may be deducted for careless or badly arranged work.

Questions 1 – 5 (5 marks)

Write the letter corresponding to the correct answer on your answer sheet.

1 What are the x -coordinates of the two turning points to the curve $f(x) = x^3 - 12x^2 + 36x + 10$?

- (A) $x = -2, x = -6$
- (B) $x = 2, x = 6$
- (C) $x = 0, x = 3$
- (D) $x = 3, x = 4$

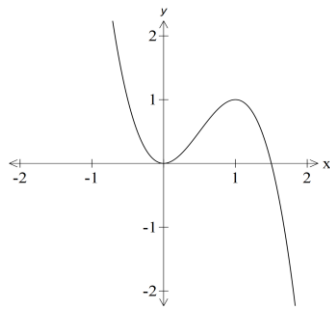
2 The graph $y = f(x)$ passes through the point $(1, 4)$ and $f'(x) = 3x^2 - 2$.

Which of the following expressions is $f(x)$?

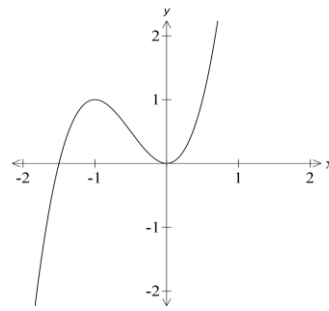
- (A) $x^3 - 2x$
- (B) $2x - 1$
- (C) $x^3 - 2x + 3$
- (D) $x^3 - 2x + 5$

3 Which of the following is the graph of $f(x) = 2x^3 - 3x^2$?

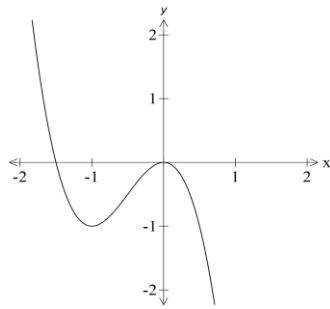
(A)



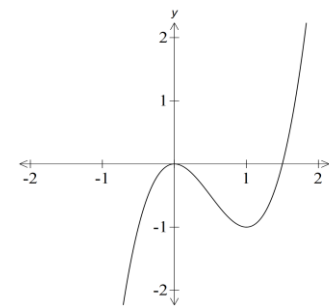
(B)



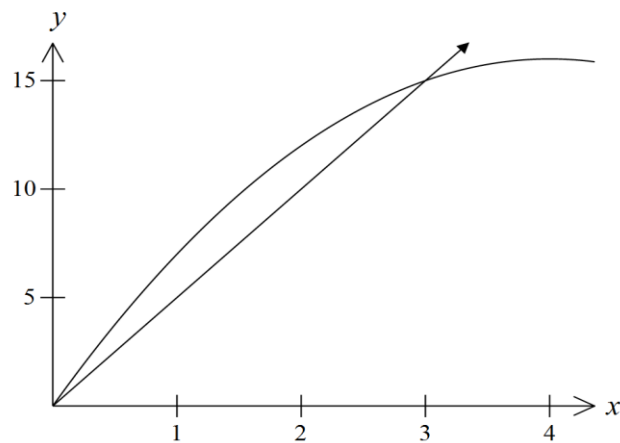
(C)



(D)



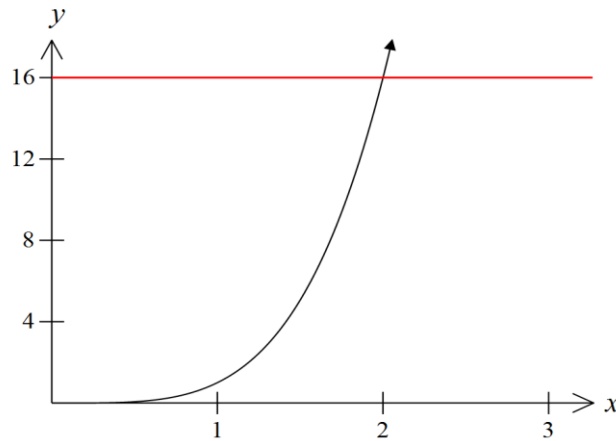
4 The diagram below shows the graph of $y = 5x$ and $y = 8x - x^2$.



What is the area between the curves $y = 5x$ and $y = 8x - x^2$?

- (A) 4.5 units²
- (B) 5.5 units²
- (C) 9.0 units²
- (D) 13.5 units²

- 5 A region in the diagram is bounded by the curve $y = x^4$, the y -axis and the line $y = 16$.



Which of the following expressions is correct for the volume of the solid of revolution when this region is rotated about the y -axis?

- (A) $V = \pi \int_0^2 x^8 dx$
 (B) $V = \pi \int_0^{16} x^8 dx$
 (C) $V = \pi \int_0^2 y^{\frac{1}{2}} dy$
 (D) $V = \pi \int_0^{16} y^{\frac{1}{2}} dy$

For Questions 6 – 12, show all working. Start each question on a new sheet of paper.

Question 6 (9 marks)

- a. The curve $y = 3x^2 + \frac{a}{x^2}$ has a turning point at $x = 3$. Find the value of a . 3
- b. If $y = 2x\sqrt{x}$, show that $\frac{y'}{y''} = 2x$. 3
- c. For what values of x is the curve $y = x^4 - 4x^3 - 18x^2$ concave up? 3

Question 7 (13 marks)

- a. Find the second derivative of $\frac{2x+1}{2x-1}$. 4
- b. For the function $f(x) = x^4 - 4x^3$,
- (i) Find the stationary points on the curve and determine their nature. 4
 - (ii) Find any points of inflexion. 2
 - (iii) Sketch the curve, showing all important features including the intercepts. 3

Question 8 (11 marks)

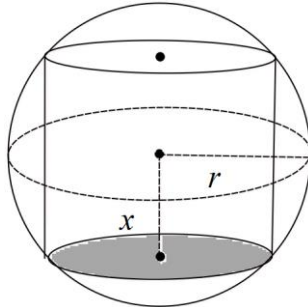
- a. The cost per hour of a bike ride is given by the formula

$$C = x^2 - 15x + 70 \text{ where } x \text{ is the distance travelled in kilometres.}$$

Find, using calculus, the distance that gives the minimum cost.

4

- b. A cylinder is to be made to fit inside a sphere of radius r cm as shown.



Let x be the distance of the base of the cylinder from the centre of the sphere.

- (i) Find an expression for the radius of the base of the cylinder in terms of r and x .
 (ii) Show that the volume, V , of the cylinder is given by

1

$$V = 2\pi x(r^2 - x^2)$$

2

- (iii) Find, in terms of r , the maximum volume of the cylinder.

5

Question 9 (13 marks)

- a. Find the primitive of:

(i) $\frac{x^4 - 3x + 4}{x^3}$

3

(ii) $\frac{1}{\sqrt{2x+3}}$

3

- b. The gradient function of a curve is $\frac{dy}{dx} = 3 - \frac{2}{x^2}$.

Find the equation of the curve if it passes through the point $(1, -2)$.

3

- c. Find the equation of the curve $y = f(x)$, given that $\frac{d^2y}{dx^2} = 2x + 3$ and there is a minimum at $(1, 3)$.

4

Question 10 (17 marks)

a. Find:

(i) $\int (x^3 - 5x^2 + 7)dx$ 3

(ii) $\int \left(\frac{1}{x^2} + \frac{2}{\sqrt{x}} \right) dx$ 3

(iii) $\int x^2(x^2 + 3x - 4)dx$ 3

b. Evaluate:

(i) $\int_{\frac{1}{9}}^{\frac{1}{3}} \frac{t+5}{\sqrt{t^3}} dt$ 4

(ii) $\int_{-1}^1 (3x + 2)^3 dx$ 4

Question 11 (9 marks)

a. Let $f(x) = \sqrt{25 - x^2}$.

(i) Copy and complete the table of values. 1

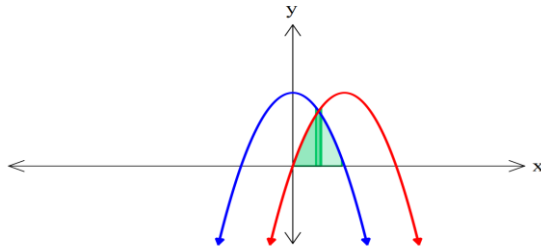
x	0	1	2	3	4	5
$f(x)$						

(ii) Use the Trapezoidal Rule with these function values to find an approximation for $\int_0^5 \sqrt{25 - x^2} dx$ correct to 3 decimal places. 3

b. Use 2 applications of Simpson's Rule to find an approximate value for the area bounded by the curve $y = \frac{4}{x}$, the x -axis and the lines $x = 1$ and $x = 2$. 5

Question 12 (13 marks)

- a. Find the area between the curve $y = x^2 - 5x + 4$ and the x - axis between $x = 2$ and $x = 5$. 4
- b. The shaded area is enclosed between the parabolas $f(x) = 4 - x^2, g(x) = -x^2 + 4x$ and the x - axis. Find the shaded area. 4



- c. The area enclosed between the parabola $f(x) = x^2 + 2$ and the line $g(x) = 4x - 1$ is rotated about the y - axis. Find the volume of the solid generated. 5

End of Examination

YEAR 12 MATHEMATICS TASK 2 2015
SOLUTIONS

MC (5 marks)

1. $f(x) = x^3 - 12x^2 + 36x + 10$

$f'(x) = 3x^2 - 24x + 36$

TP $\Rightarrow f'(x) = 0$

$3(x^2 - 8x + 12) = 0$

$(x-2)(x-6) = 0$

$x = 2, 6$

B

2. $f'(x) = 3x^2 - 2$

$f(x) = x^3 - 2x + C$

passes through (1, 4)

i.e. $1^3 - 2(1) + C = 4$

$C = 5$

$\therefore f(x) = x^3 - 2x + 5$

D

3. $f(x) = 2x^3 - 3x^2$

x-int $\Rightarrow x^2(2x-3) = 0$

$x = 0, \frac{3}{2}$

double roots

SP $\Rightarrow f'(x) = 6x^2 - 6x = 0$

$6x(x-1) = 0$

$x = 0, 1$

$f''(x) = 12x - 6$

At (0, 0), $f''(x) = -6 \therefore \text{max}$

At (1, -1), $f''(x) = 6 \therefore \text{min}$

D

4. $A = \int_0^3 (8x - x^2 - 5x) dx$

$= \int_0^3 (3x - x^2) dx$

$= \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_0^3$

$= 4\frac{1}{2}$

A

5. $y = x^4 \Rightarrow x^2 = y^{\frac{1}{2}}$

$V = \pi \int_0^4 y^{\frac{1}{2}} dy$

D

Question 6 (9 marks)

a) $y = 3x^2 + \frac{a}{x^2}$
 $= 3x^2 + ax^{-2}$

$\frac{dy}{dx} = 6x - \frac{2a}{x^3}$

TP $\Rightarrow \frac{dy}{dx} = 0$

$6(3) - \frac{2a}{3^3} = 0$

$\frac{2a}{27} = 18$

$a = 243$

3

b) $y = 2x\sqrt{x}$
 $= 2x^{\frac{3}{2}}$

$\frac{dy}{dx} = 3x^{\frac{1}{2}} ; \frac{d^2y}{dx^2} = \frac{3}{2}x^{-\frac{1}{2}}$
 $= 3\sqrt{x} \quad \quad \quad = \frac{3}{2\sqrt{x}}$

$\frac{y'}{y''} = 3\sqrt{x} \times \frac{2\sqrt{x}}{3}$

$= 2x$

3

c) $y = x^4 - 4x^3 - 18x^2$

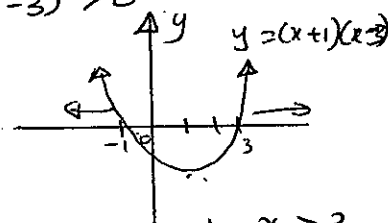
$\frac{dy}{dx} = 4x^3 - 12x^2 - 36x$

$\frac{d^2y}{dx^2} = 12x^2 - 24x - 36$

Concave up $\Rightarrow \frac{d^2y}{dx^2} > 0$

$12(x^2 - 2x - 3) > 0$

$(x+1)(x-3) > 0$



Concave up when $x < -1, x > 3$

3

Question 7 (13 marks)

a) $y = \frac{2x+1}{2x-1}$

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$$

$$= \frac{2(2x-1) - 2(2x+1)}{(2x-1)^2}$$

$$= \frac{-4}{(2x-1)^2} = -4(2x-1)^{-2}$$

$$\frac{d^2y}{dx^2} = 8(2x-1)^{-3} \cdot 2$$

$$= \frac{16}{(2x-1)^3}$$

(4)

b) $f(x) = x^4 - 4x^3$

$$f'(x) = 4x^3 - 12x^2; f''(x) = 12x^2 - 24x$$

i) SP $\Rightarrow f'(x) = 0$

$$4x^3 - 12x^2 = 0$$

$$4x^2(x-3) = 0$$

$$x = 0, 3$$

when $x=3$, $f''(3) = 12(3)^2 - 24(3) = 36 > 0$

$\therefore$ minimum at $(3, -27)$

when $x=0$, $f''(0) = 0$

$\therefore$ possible point of inflexion

Test for Point of Inflexion

x	-1	0	1
f''(x)	36	0	-12

there is a change in concavity

$\therefore$ Horizontal point of inflexion at $(0, 0)$ (4)

ii) Points of inflexion $\Rightarrow f''(x) = 0$

$$12x^2 - 24x = 0$$

$$12x(x-2) = 0$$

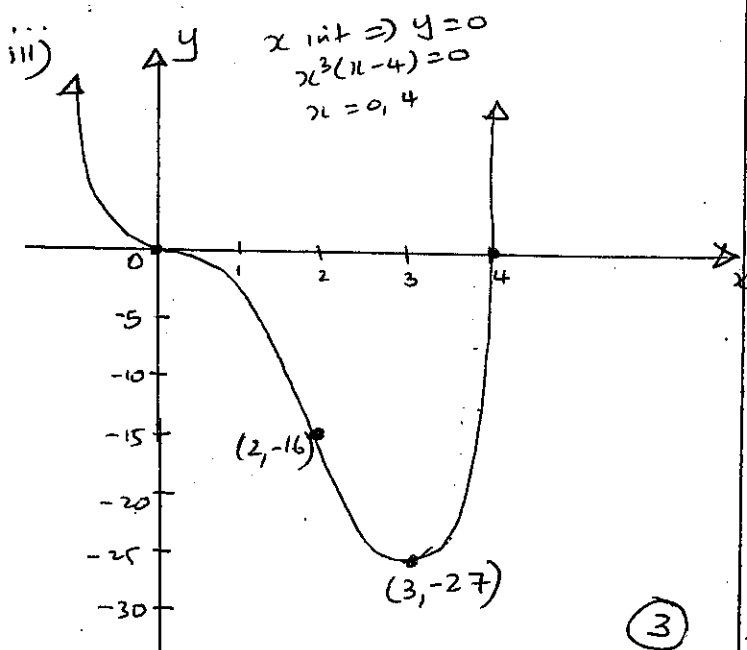
$$x = 0, x = 2$$

Horizontal point of inflexion at $(0, 0)$ — shown in (i)

Test $x=2$

x	1.5	2	2.5
f''(x)	-9	0	15

there is a change in concavity
 $\therefore$ point of inflexion at $(2, -16)$ (2)



Question 8 (11 marks)

a) $C = x^2 - 15x + 70$

$$\frac{dC}{dx} = 2x - 15; \frac{d^2C}{dx^2} = 2$$

For minimum cost, $\frac{dC}{dx} = 0$

$$\text{i.e. } 2x - 15 = 0$$

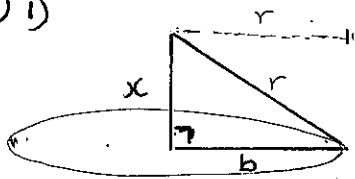
$$x = 7\frac{1}{2}$$

When $x = 7\frac{1}{2}$, $\frac{d^2C}{dx^2} > 0$

$\therefore$ Minimum when $x = 7\frac{1}{2} \text{ km}$ (4)

8. let radius of base of cylinder = b

b) i)



$$b^2 = r^2 - x^2$$

$$b = \sqrt{r^2 - x^2} \quad (1)$$

ii) Height of cylinder = $2x$

$$V = \pi r^2 h$$

$$= \pi (\sqrt{r^2 - x^2})^2 (2x)$$

$$= 2\pi x (r^2 - x^2) \quad (2)$$

$$\text{iii) } V = 2\pi r^2 x - 2\pi x^3$$

$$\frac{dV}{dx} = 2\pi r^2 - 6\pi x^2$$

maximum V when $\frac{dV}{dx} = 0$

$$2\pi r^2 - 6\pi x^2 = 0$$

$$2\pi (r^2 - 3x^2) = 0$$

$$r^2 = 3x^2$$

$$x = \frac{r}{\sqrt{3}} \quad (\text{since } x > 0)$$

$$\frac{d^2V}{dx^2} = -12\pi x$$

$$\text{When } x = \frac{r}{\sqrt{3}}, \quad \frac{d^2V}{dx^2} = \frac{-12\pi r}{\sqrt{3}}$$

< 0

$\therefore$ maximum when $x = \frac{r}{\sqrt{3}}$

Maximum Volume

$$= 2\pi \frac{r}{\sqrt{3}} \left(r^2 - \frac{r^2}{3} \right)$$

$$= \frac{4\pi r^3}{3\sqrt{3}} \text{ cm}^3 \quad (4)$$

Question 9 (13 marks)

$$\text{a) i) } y' = \frac{x^4 - 3x + 4}{x^3}$$

$$= x - \frac{3}{x^2} + \frac{4}{x^3}$$

$$= x - 3x^{-2} + 4x^{-3}$$

$$y = \frac{x^2}{2} + 3x^{-1} - \frac{4x^{-2}}{2} + C$$

$$= \frac{x^2}{2} + \frac{3}{x} - \frac{2}{x^2} + C \quad (3)$$

$$\text{ii) } y' = \frac{1}{\sqrt{2x+3}}^{-1/2}$$

$$= (2x+3)^{-1/2}$$

$$y = \frac{(2x+3)^{1/2}}{\frac{1}{2} \times 2} + C$$

$$y = \sqrt{2x+3} + C \quad (3)$$

$$\text{b) } \frac{dy}{dx} = 3 - \frac{2}{x^2}$$

$$= 3 - 2x^{-2}$$

$$y = 3x + 2x^{-1} + C$$

$$y = 3x + \frac{2}{x} + C$$

When $x=1, y=-2$

$$-2 = 3(1) + \frac{2}{1} + C$$

$$C = -7$$

$$\therefore y = 3x + \frac{2}{x} - 7 \quad (3)$$

$$\text{c) } \frac{d^2y}{dx^2} = 2x + 3$$

$$\frac{dy}{dx} = x^2 + 3x + C$$

Minimum at $(1, 3)$

ie. when $x=1, \frac{dy}{dx} = 0$

$$(1)^2 + 3(1) + C = 0$$

$$C = -4$$

$$\therefore \frac{dy}{dx} = x^2 + 3x - 4$$

9) cont.

$$c) y = \frac{x^3}{3} + \frac{3x^2}{2} - 4x + c$$

passes through (1,3)

$$\therefore \frac{1}{3} + \frac{3}{2} - 4 + c = 3$$

$$c = 5\frac{1}{6}$$

$$\therefore y = \frac{x^3}{3} + \frac{3x^2}{2} - 4x + 5\frac{1}{6}$$

(4)

Question 10 (17 marks)

$$a) i) \int (x^3 - 5x^2 + 7) dx$$

$$= \frac{x^4}{4} - \frac{5x^3}{3} + 7x + C$$

(3)

$$ii) \int \left(\frac{1}{x^2} + \frac{2}{\sqrt{x}} \right) dx$$

$$= \int (x^{-2} + 2x^{-1/2}) dx$$

$$= -x^{-1} + 4x^{1/2} + C$$

$$= -\frac{1}{x} + 4\sqrt{x} + C$$

(3)

$$iii) \int x^2(x^2 + 3x - 4) dx$$

$$= \int (x^4 + 3x^3 - 4x^2) dx$$

$$= \frac{x^5}{5} + \frac{3x^4}{4} - \frac{4x^3}{3} + C$$

(3)

$$b) i) \int_1^9 \frac{t+5}{\sqrt{t^3}} dt$$

$$= \int_1^9 \frac{t+5}{t^{3/2}} dt$$

$$= \int_1^9 (t^{-1/2} + 5t^{-3/2}) dt$$

$$= \left[2\sqrt{t} - \frac{10}{\sqrt{t}} \right]_1^9$$

$$= 6 - \frac{10}{3} - (-8)$$

$$= 10\frac{2}{3}$$

(4)

$$ii) \int_{-1}^0 (3x+2)^3 dx$$

$$= \left[\frac{(3x+2)^4}{12} \right]_{-1}^0$$

$$= \frac{1}{12} (2^4 - 1^4)$$

$$= \frac{15}{12}$$

$$= \frac{5}{4}$$

(4)

Question 11 (9 marks)

a) i) $f(x) = \sqrt{25-x^2}$

	1	2	2	2	2	1
x	0	1	2	3	4	5
$f(x)$	5	4.899	4.583	4	3	0

$h=1$

(1)

ii) $\int_0^5 \sqrt{25-x^2} dx \doteq \frac{1}{2} \left\{ 5 + 2(4.899 + 4.583 + 4 + 3) + 0 \right\}$

$\doteq 18.982$

(3)

b)

	1	4	2	4	1
x	1	$1\frac{1}{4}$	$1\frac{1}{2}$	$1\frac{3}{4}$	2
$f(x)$	4	$\frac{16}{5}$	$\frac{8}{3}$	$\frac{16}{7}$	2

$h=0.25$

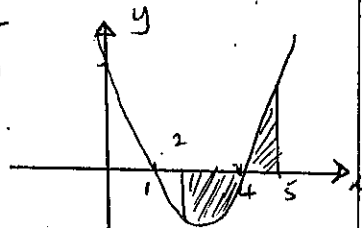
$A \doteq \frac{0.25}{3} \left\{ 4 + 4\left(\frac{16}{5} + \frac{16}{7}\right) + 2\left(\frac{8}{3}\right) + 2 \right\}$

$\doteq 2 \frac{487}{630}$

(5)

Question 12 (13 marks)

a) $y = x^2 - 5x + 4$
 $= (x-1)(x-4)$



$$A = \left| \int_1^4 (x^2 - 5x + 4) dx \right| + \int_4^5 (x^2 - 5x + 4) dx$$

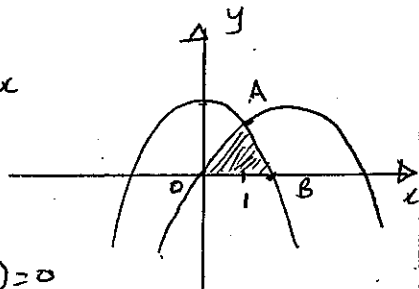
$$= \left| \left[\frac{x^3}{3} - \frac{5x^2}{2} + 4x \right]_1^4 \right| + \left[\frac{x^3}{3} - \frac{5x^2}{2} + 4x \right]_4^5$$

$$= \left| \left(\frac{4^3}{3} - \frac{5(4)^2}{2} + 4(4) \right) - \left(\frac{1^3}{3} - \frac{5(1)^2}{2} + 4(1) \right) \right| + \left(\frac{5^3}{3} - \frac{5(5)^2}{2} + 4(5) \right) - \left(\frac{4^3}{3} - \frac{5(4)^2}{2} + 4(4) \right)$$

$$= 3\frac{1}{3} + 1\frac{5}{6}$$

$$= 5\frac{1}{6} \text{ square units} \quad (4)$$

b) $A: 4 - x^2 = -x^2 + 4x$
 $x = 1$



B: $f(x) = 4 - x^2$
 $x \text{ int} \Rightarrow (2-x)(2+x) = 0$
 $x = \pm 2$
 $\therefore B(2, 0)$

$$A = \int_0^1 (4x - x^2) dx + \int_1^2 (4 - x^2) dx$$

$$= \left[2x^2 - \frac{x^3}{3} \right]_0^1 + \left[4x - \frac{x^3}{3} \right]_1^2$$

$$= \left(2 - \frac{1}{3} - 0 \right) + \left(\left(8 - \frac{8}{3} \right) - \left(4 - \frac{1}{3} \right) \right)$$

$$= \frac{5}{3} + \frac{5}{3}$$

$$= 3\frac{1}{3} \text{ square units}$$

c) $f(x) = x^2 + 2$; $g(x) = 4x - 1$

Points of Intersection

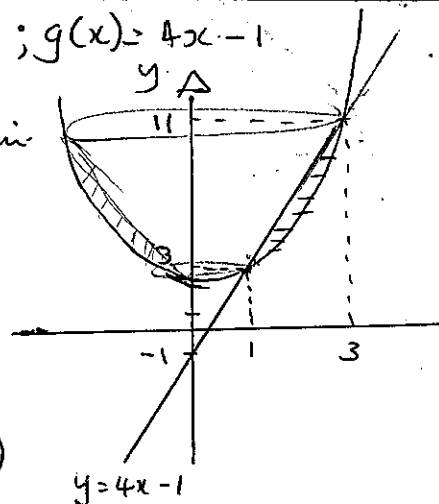
$$x^2 + 2 = 4x - 1$$

$$x^2 - 4x + 3 = 0$$

$$(x-1)(x-3) = 0$$

$$x = 1, 3$$

$$\text{pts } (1, 3), (3, 11)$$



$$y = x^2 + 2$$

$$x^2 = y - 2$$

$$y = 4x - 1$$

$$x = \frac{y+1}{4}$$

$$x^2 = \frac{(y+1)^2}{16}$$

$$V = \pi \int_3^{11} (y-2) dy - \pi \int_3^{11} \frac{(y+1)^2}{16} dy$$

$$= \pi \int_3^{11} \left(y - 2 - \frac{(y+1)^2}{16} \right) dy$$

$$= \pi \left[\frac{y^2}{2} - 2y - \frac{(y+1)^3}{48} \right]_3^{11}$$

$$= \pi \left[\left(\frac{11^2}{2} - 2(11) - \frac{12^3}{48} \right) - \left(\frac{3^2}{2} - 2(3) - \frac{4^3}{48} \right) \right]$$

$$= \frac{16\pi}{3} \text{ cubic units}$$

(5)